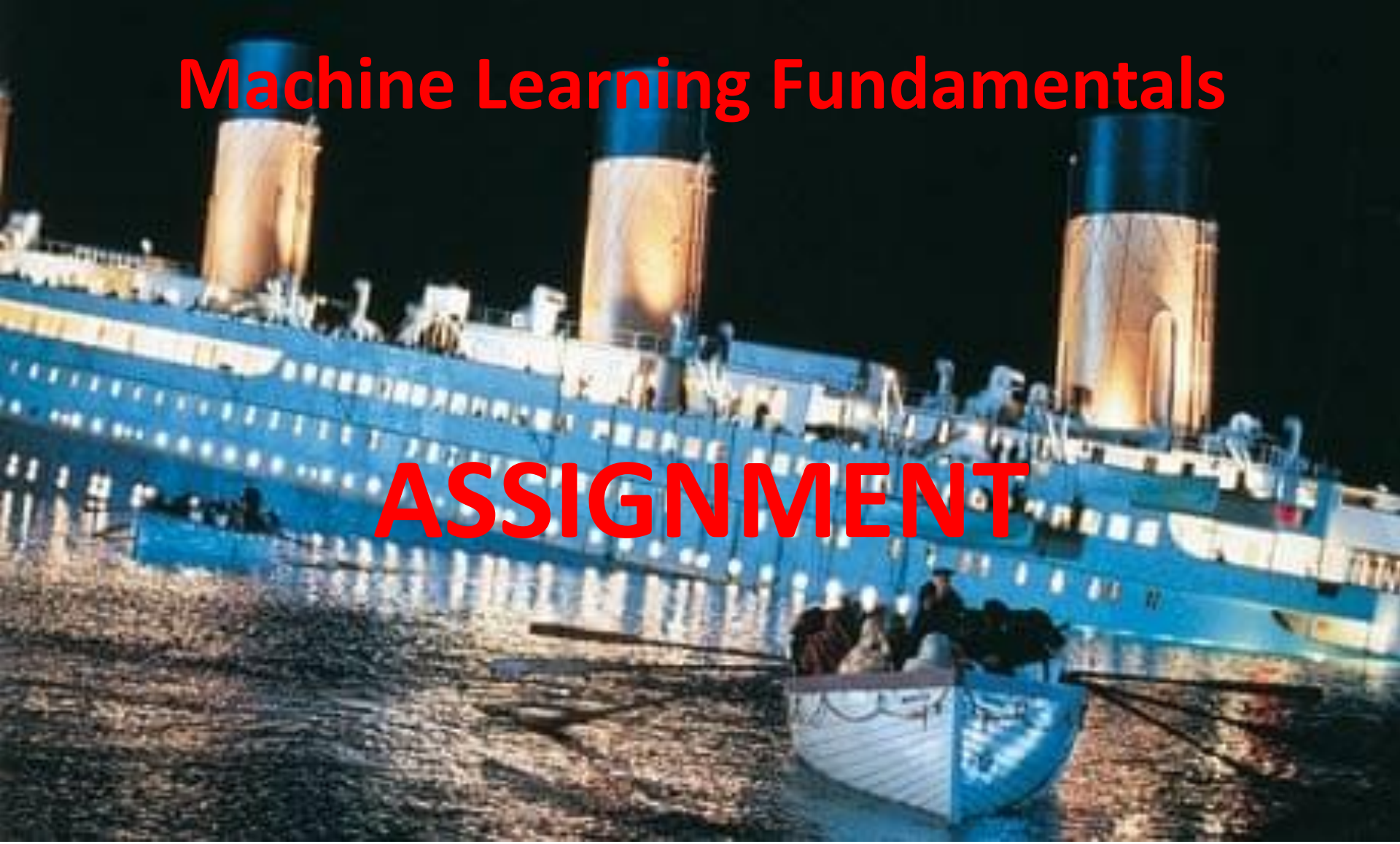
Machine Learning Fundamentals



ASSIGNMENT

Andy Louwyck

Academic year 2025 - 2026

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Participate in the Titanic Competition

kaggle.com/competitions/titanic/overview

Submit Prediction

Search

KAGGLE · GETTING STARTED PREDICTION COMPETITION · ONGOING

Titanic - Machine Learning from Disaster

Start here! Predict survival on the Titanic and get familiar with ML basics

Overview | Data | Code | Models | Discussion | Leaderboard | Rules | Team | Submissions

Instructions

- Clean the data and perform an **exploratory data analysis** (EDA)

- Use **shallow learning models** (e.g. ensemble methods)
- Do not use deep learning models (a.k.a. neural networks)

- Code in **Python** and use **Scikit-Learn** and related packages such as NumPy, Pandas, SciPy, Matplotlib, and Seaborn

- **Mention the sources** you consulted and the AI tools you used

- Preferred language is **English**.
  (Dutch is also an option.)

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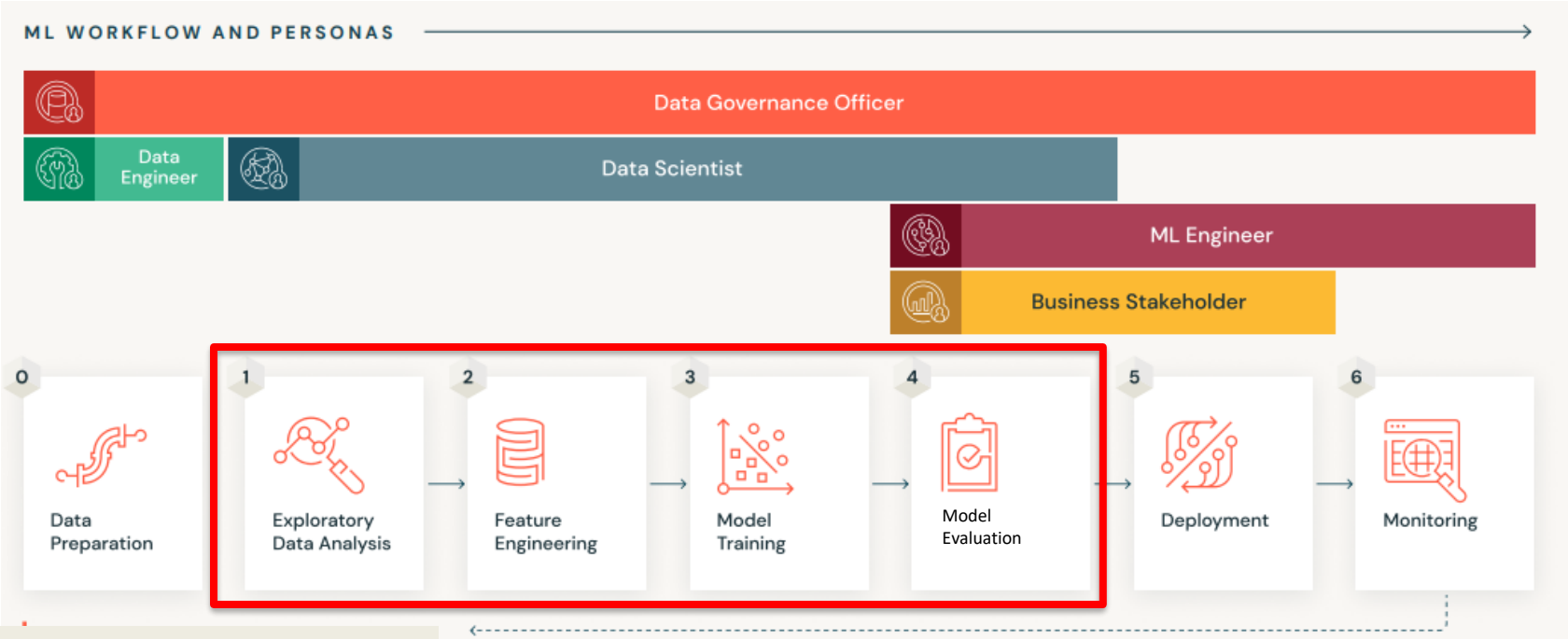
Machine Learning Workflow



The Big Book of MLops (Databricks, 2023)

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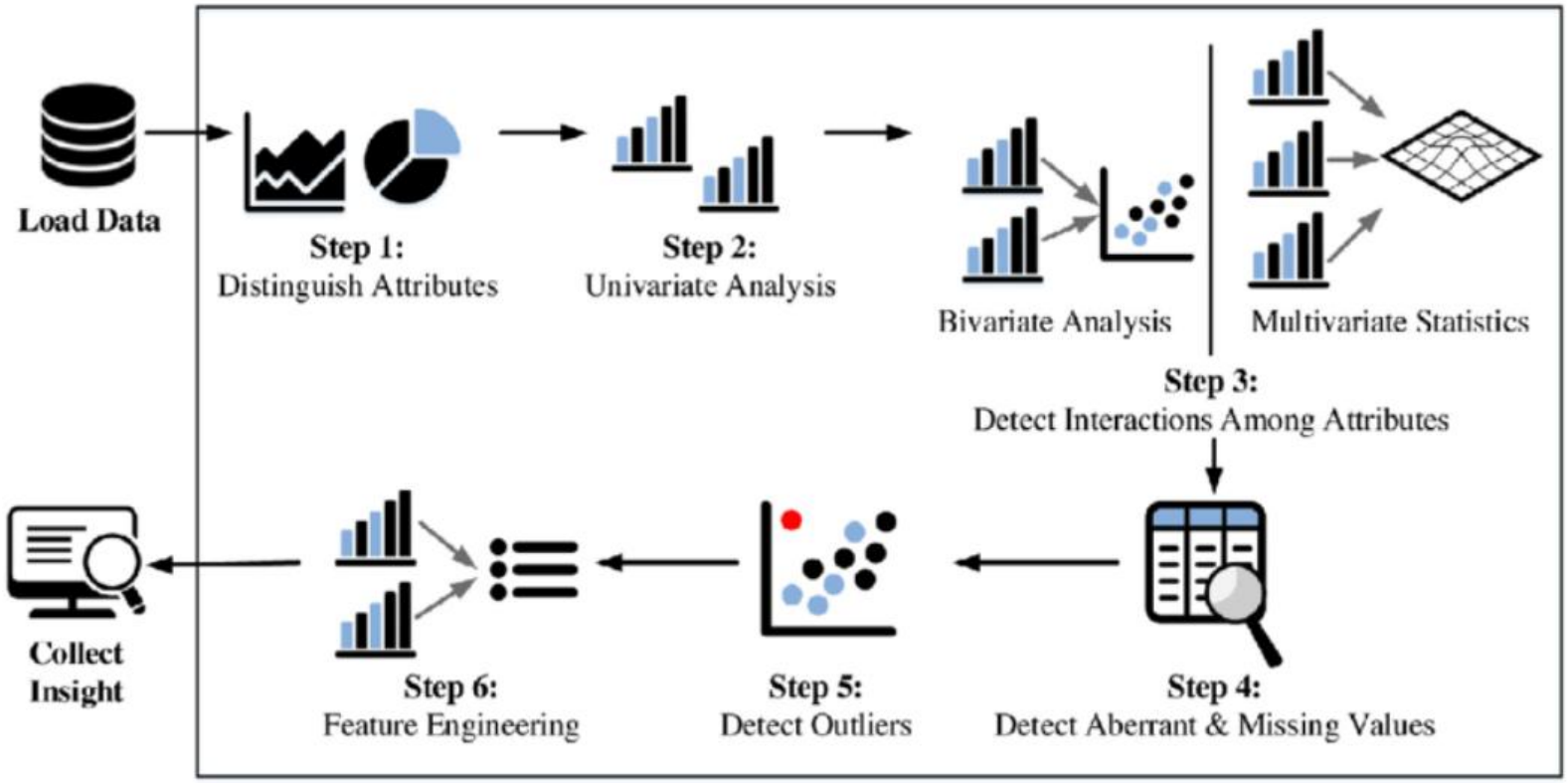
Exploratory Data Analysis



Source: https://doi.org/10.1016/j.visinf.2018.12.004

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Feature Normalization and Encoding

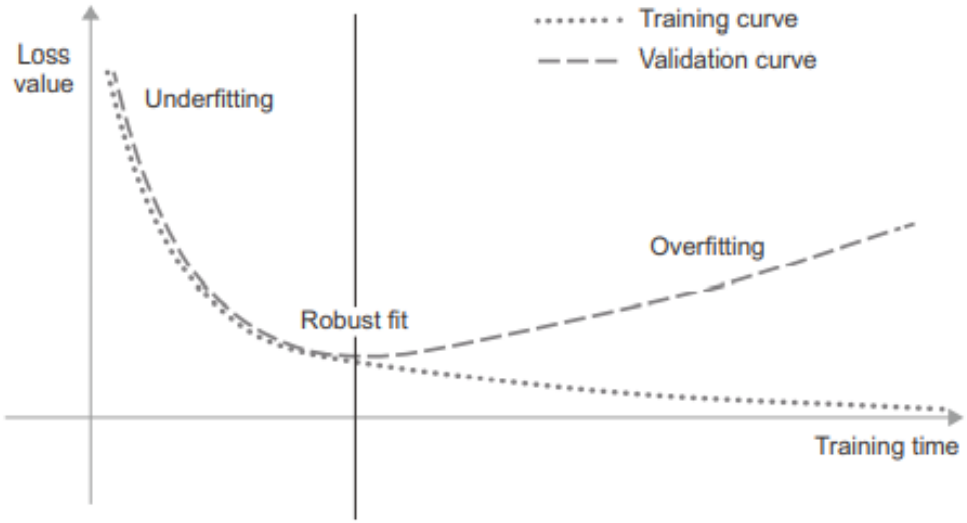
```
from sklearn.preprocessing import MinMaxScaler, StandardScaler
from sklearn.preprocessing import LabelEncoder, OrdinalEncoder, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
```

- **Quantitative variables:**
  - Scaling: MinMaxScaler
  - Standardization: StandardScaler
  - …
- **Categorical variables:**
  - Label encoding: LabelEncoder
  - Ordinal encoding: OrdinalEncoder
  - One-hot encoding: OneHotEncoder
- **Combining steps:**
  - ColumnTransformer
  - Pipeline

Optimization vs Generalization

- **<u>Optimization</u>:** fitting the data as best as possible during <u>training</u>

- **<u>Generalization</u>:** good model performance on new data during <u>inference</u>



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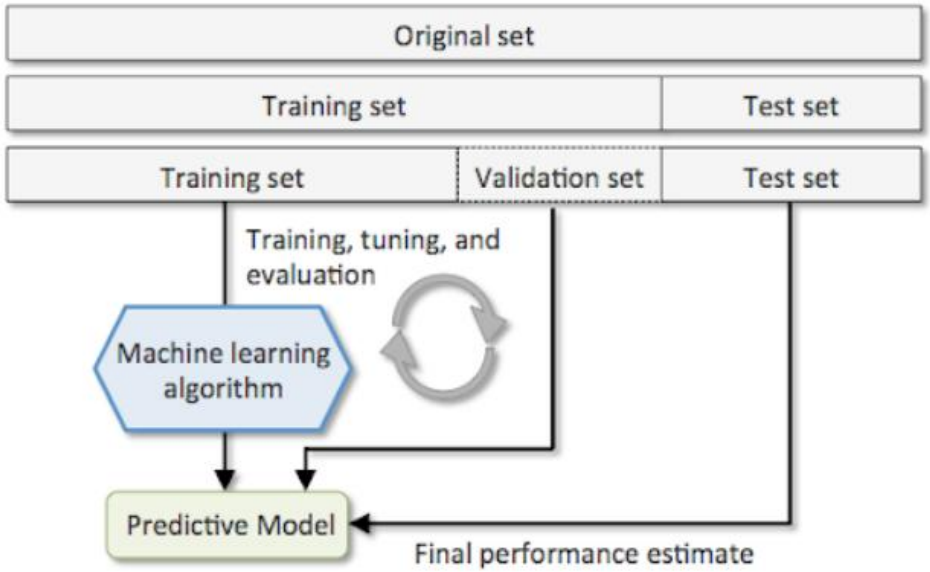
Training, Validating, and Testing

```
from sklearn.model_selection import train_test_split
```

Splitting the dataset:

- **training**:
  - deriving the optimal model parameters
  - by the machine learning algorithm
- **validating**:
  - finding the optimal model configuration
  - fine-tuning the hyperparameters
  - to overcome overfitting
  - by the machine learning engineer
- **testing**:
  - final evaluation
  - by the machine learning engineer



(source: https://vitalflux.com/hold-out-method-for-training-machine-learning-model)

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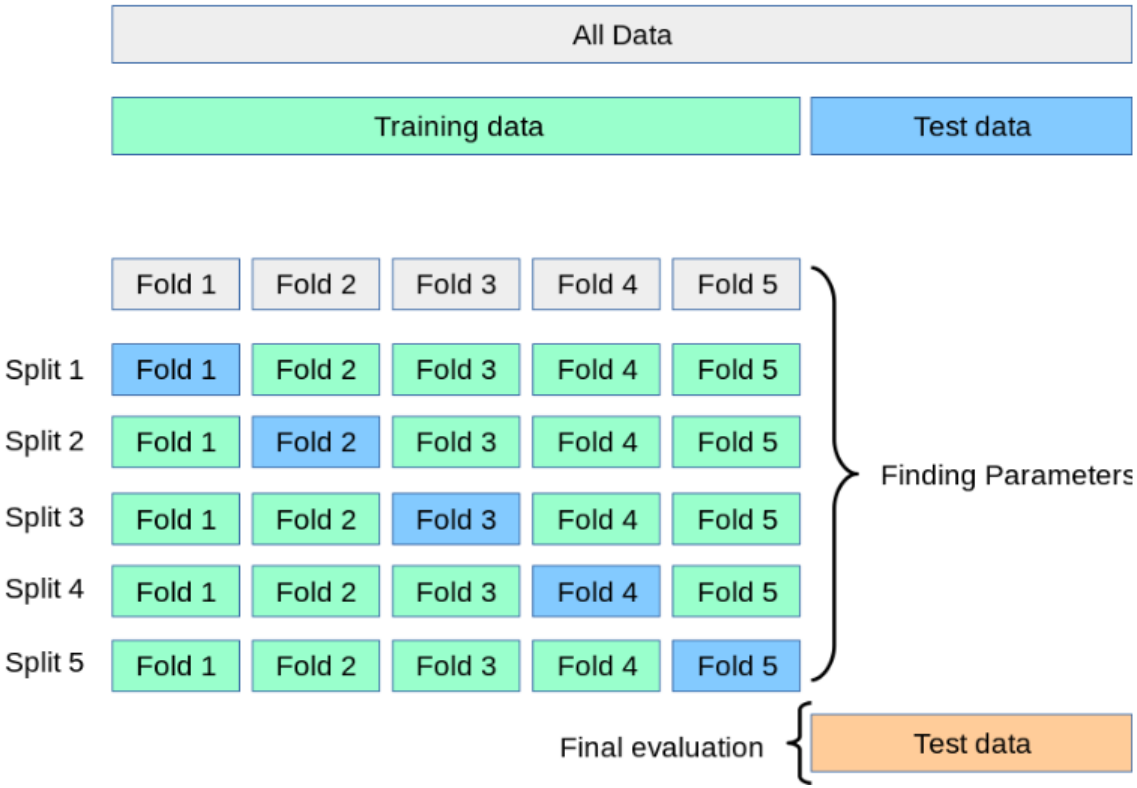
Cross Validation

```
from sklearn.model_selection import GridSearchCV
```

Recommended in case of **small datasets!**

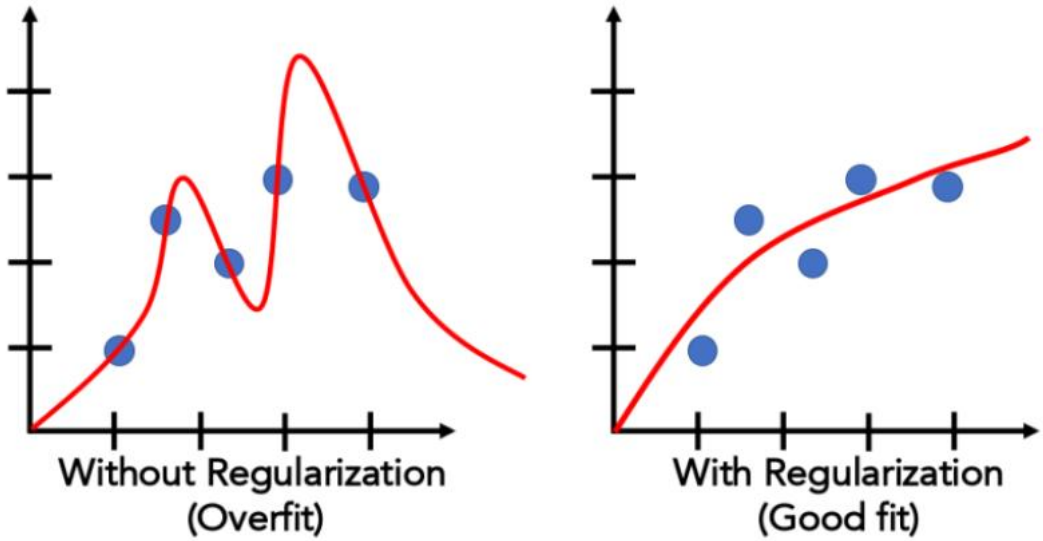


Source: https://scikit-learn.org/stable/modules/cross_validation.html

Regularization

= preventing overfitting by adding constraints or penalties to the optimization process

- LASSO = L1 regularization
- RIDGE = L2 regularization
- Elastic Net = L1 + L2
- Dropout in neural nets
- Early stopping



https://www.linkedin.com/pulse/lasso-vs-ridge-regression-feature-selection-mouhssine-akkouh-diiye/

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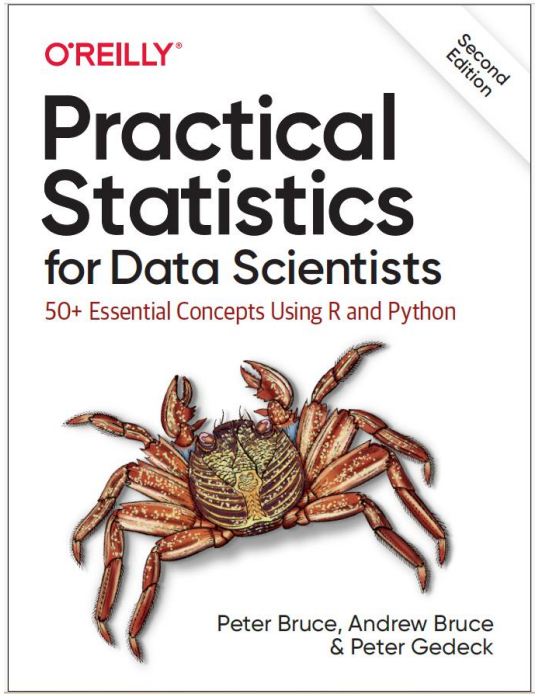
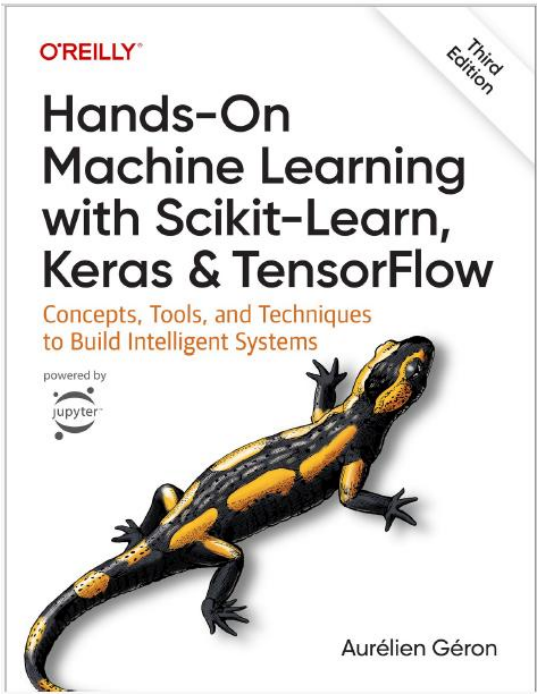
Useful Tips

Chapter 1: EDA



Chapter 7: Ensemble Learning



= learning material!

Deliverables

- **Proof** (e.g., screenshot) that you participated in the competition and the *score* you achieved.
- A Jupyter **notebook** in the style of a data science blog that explains step-by-step how you arrived at your final model. The *reading time* should not exceed *15 minutes*.
- A **video** explaining your solution method. Be sure to explain how your final model works and the training algorithm you applied to optimize the model parameters. The video should be at least *10 minutes* and no more than *15 minutes* long. Make sure *you are visible* in the video.
- Make sure that **any link** to your video or notebook is and **remains** permanently **accessible** to the course instructor!

- **Deadline: Sunday, May 24** (midnight)

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Proof of Submission

Search

Andy Louwyck

Overview Data Code Models Discussion Leaderboard Rules Team Submissions

Submissions

All Successful Errors

Recent

Submission and Description	Public Score
submission.csv Complete · 32s ago · test CSV file for submission (all predictions are zero)	**0.62200**

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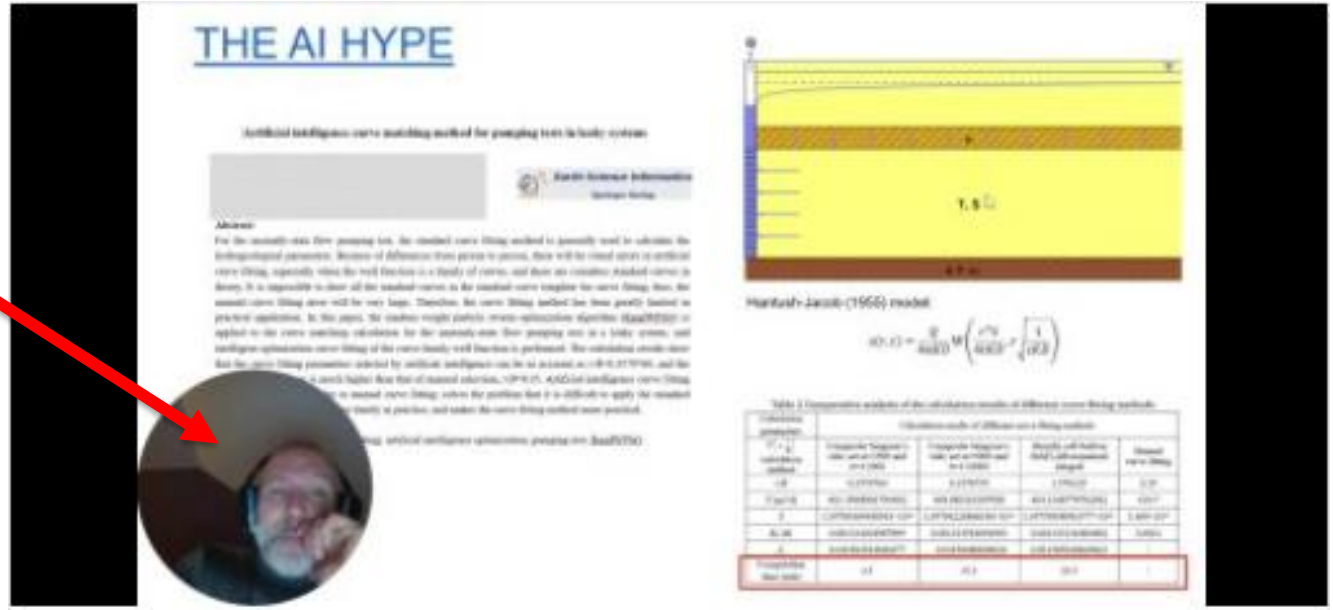
Video

Microsoft Stream



Introduction

U hebt dit bewerkt

Submit in Toledo

Assignment

Machine_Learning_Fundamentals_Assignment.pdf

Kaggle Competition: Titanic - Machine Learning from Disaster

Submission Titanic

Due date: 5/24/26, 11:59 PM (UTC+2)

Titanic assignment.pdf

Disclaimer

The submitted material will be treated confidentially and will not be further distributed.

Only the course instructor will review and examine the material.

In exceptional cases, the material may also be reviewed by a colleague instructor.

The course instructor may request additional clarification in case of suspected fraud.

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ECTS (daytime education)

Assessment activities

Machine Learning - Fundamentals (B-VIV-V82975)

Type: Partial or continuous assessment with (final) exam during the examination period
Assessment method: Oral, Paper/Project, Presentation, Project/Product, Written
Type of questions: Multiple choice, Open questions, Closed questions
Permitted materials: Course material, Computer

Explanation | Information about resitting exams

Exam time	Grading scale
TOTAL	1-20 Grade scale
V5R312 - Machine Learning - Fundamentals	1-20 Grade scale

- Assignments during the lecture weeks: 20%
- Final exam: 80%

Challenges: 5%

Assignment: 15%

ECTS (distance education)

Evaluatieactiviteiten

Machine Learning - Fundamentals (B-VIV-V83723)

Type: Partiële of permanente evaluatie met examen tijdens de examenperiode
Evaluatievorm : Mondeling, Ontwerp/Product, Paper/Werkstuk, Presentatie, Schriftelijk, Verslag
Vraagvormen: Meerkeuzevragen, Open vragen, Gesloten vragen
Leermateriaal : Cursusmateriaal, Computer

Toelichting | **Toelichting bij herkansen**

Examenmoment	Beoordelingsschaal
TOTAAL	1-20 Puntenschaal
V5R735 - Machine Learning - Fundamentals	1-20 Puntenschaal

- Opdracht: 20%
- Eindexamen met theorievragen en oefening(en): 80%

Have Fun and Good Luck!

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